A UI/UX DESIGN WORKFLOW, MAPPED TO INDUSTRY STANDARDS

From Requirements to Interface

How SinhalaBridge's SRS → Use Cases → Pages → Content → UX → UI pipeline maps onto established design-industry process — and how to run it on the next project.

This note documents the working method used to design SinhalaBridge, and shows where each step sits inside the process frameworks most design teams already use: the **Double Diamond** (Design Council, UK), the **Design Thinking** model (Stanford d.school / IDEO), and standard **UML/requirements-engineering** practice for products with a formal SRS. The goal is a workflow that is specific enough to execute against, but recognisable to any designer or engineer who joins the project later.

FRAME OF REFERENCE

Where this fits in the Double Diamond

The Double Diamond has four phases: **Discover** (research the problem), **Define** (commit to a brief), **Develop** (explore solutions), **Deliver** (build and test). The six-step workflow below is essentially the *Define → Develop* half of that diamond, made concrete for a product that already has a locked SRS â€” which is the normal starting point in industry once discovery research (user interviews, competitive analysis) has already happened.

Discover	Define	Develop	Deliver
User research, competitive analysis	SRS, requirements, personas, acceptance criteria	Use cases → pages → content → UX → UI	Build, usability test, ship, iterate
done before this note	SinhalaBridge_SRS_v1.0	this workflow	build plan / dev handoff

THE WORKING METHOD

The Six-Step Workflow

Each step consumes the output of the one before it. Skipping a step (e.g. jumping from SRS straight to visual UI) is the most common cause of rework in industry — it produces screens that look finished but don't actually satisfy the requirement they were meant to serve.

1 SRS → Use Cases ■■■■■■■■■ → ■■■■■■ ■■■■■■■

Industry standard: IEEE 830 / UML Use Case Diagrams (behavioural modelling, before any structure is drawn).

What happens: Every functional requirement (FR-RM, FR-LC, FR-EX, FR-PG, FR-AC, FR-AD) is restated as an actor + goal + trigger. This is Phase 1 of standard UML workflow — model behaviour before architecture.

Deliverable: Use Case Diagram: user classes (Learner, Content Admin, Guest) mapped to what they can do.

Applied here: FR-RM-03 (“stages unlock sequentially”) became the use case “Learner unlocks next stage by passing the stage test.”

2 Use Cases → Pages ■■■■■■ ■■■■■■■ → ■■■■

Industry standard: Information Architecture / user-flow mapping (Sitemap + Task Flow — standard IA practice).

What happens: Use cases are grouped by the single screen where they are enacted. This produces the page inventory before any pixel is placed — the IA equivalent of a floor plan before interior design.

Deliverable: Page list with the use cases each page must support (traceability back to the SRS).

Applied here: Roadmap, Lesson, Exercise and Writing screens, each pre-mapped to specific FR IDs in SRS §5.1.

3 Pages → Content ■■■■ → ■■■■■■■■■■

Industry standard: Content-first / “content modelling” (data schema before layout — Karen McGrane's content-first principle).

What happens: For every page, define the exact data fields it needs as a schema (JSON shape), independent of how it will be styled. NFR-11 (“content separate from code”) makes this step non-negotiable, not optional.

Deliverable: Content/data model — e.g. the Stage, Lesson, Exercise and Writing objects (curriculum.json shape).

Applied here: One data.js content layer was written once and reused, unstyled, across three different visual concepts.

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Content → UX ■■■■■■■■■ → UX

Industry standard: Low-fidelity wireframing + interaction/state design (Nielsen Norman Group: structure and flow before visual design).

What happens: Define flow, states (empty / loading / error / locked / in-progress / completed) and interaction logic — grey-box level, no colour or type decisions yet.

Deliverable: Wireframes or annotated flow diagrams; state charts for anything with a lifecycle.

Applied here: Locked → in-progress → completed stage states; scaffold-then-fade flow from Stage 6 to Stage 7.

5

UX → UI UX → UI

Industry standard: Design System / visual design application (tokens → components → screens — Atomic Design, Brad Frost).

What happens: Only now are the locked visual tokens applied: palette, typography, spacing, and imagery — on top of a UX that already works without them.

Deliverable: High-fidelity UI screens / prototype, built from a token system (colour, type, layout, signature).

Applied here: Saffron/gold/teal palette, ink-on-cream & white-on-teal contrast pairs, bilingual labels, 360px mobile-first.

6

Validate & Iterate ■■■■■■■ ■■ ■■■■■■■

Industry standard: Usability testing + heuristic evaluation (Nielsen's 10 Usability Heuristics; WCAG 2.1 AA for accessibility).

What happens: Test the built screens against real users or heuristics, check contrast/accessibility compliance, and feed findings back into steps 3–5 rather than starting over.

Deliverable: Usability findings, a contrast/accessibility audit, and a prioritised revision list.

Applied here: WCAG AA contrast audit run on the palette before any component was built, not after.

CROSS-REFERENCE

Workflow vs. Common Industry Frameworks

The six steps aren't a bespoke invention — they're a specific sequencing of frameworks most teams already use in some form. Naming the equivalence makes it easier to onboard a collaborator or explain the process to a stakeholder.

This workflow	Double Diamond	Design Thinking	UML phase
SRS → Use Cases	Define	Define	Phase 1 — High-level analysis (behavioural)
Use Cases → Pages	Define / Develop	Define	Phase 2 — High-level architecture (structural)
Pages → Content	Develop	Ideate	Phase 3 — Detailed design (data model)
Content → UX	Develop	Prototype	Phase 3 — State machine / sequence diagrams
UX → UI	Develop / Deliver	Prototype	— (visual design layer, outside UML)
Validate & Iterate	Deliver	Test	— (QA / acceptance testing)

HANDOFF CHECKLIST

What “done” looks like at each step

— **Use Cases:** every Must-priority FR traces to at least one actor + goal statement.

— **Pages:** every page traces back to the use cases it satisfies (no orphan screens, no uncovered use cases).

— **Content:** a versioned schema (JSON or equivalent) exists independent of any styling.

— **UX:** every stateful element (locked/unlocked, pass/fail, empty/error) is drawn, not implied.

— **UI:** a token system (colour, type, layout, signature) is defined once and applied consistently — not decided screen-by-screen.

— **Validation:** contrast ratios checked against WCAG 2.1 AA before build; usability check-in scheduled after build.

APPLIED TO SINHALABRIDGE

This week's output, in workflow terms

The three UI concepts delivered this week (Rasa, Nimnaya, Lipi) are three different answers to **Step 5 (UX → UI)** only — they share the exact same Steps 1–4: the same SRS, the same use cases, the same four pages (Roadmap, Lesson, Exercise, Writing), and the same content schema (`data.js`). This is deliberate: it isolates the one decision still open (visual language) from the decisions already locked (what the product does and how it behaves), so picking a direction doesn't require re-deciding anything upstream.

> ***Next step in the sequence:*** *Step 6 — pick one concept (or a hybrid), run a WCAG contrast pass on the final combination, and move into build against the existing 3-week Spring Boot / React plan.*